

Meta Ads Performance Analysis - Final Project Report

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Project Type: Data Analytics | Power BI Dashboard

Reference Document: Business Requirements Document (BRD)

1. Project Overview

This project presents a comprehensive Meta Ads Performance Analysis Dashboard developed to evaluate and optimize paid advertising campaigns across Facebook and Instagram.

The solution adheres to the requirements outlined in the Business Requirements Document and extends beyond them by integrating advanced KPIs, deeper demographic insights, time-based analyses, and platform-level comparisons.

The dashboard transforms raw Meta Ads data into a structured, business-ready analytical system.

2. Business Objective

The objectives of this analytical solution were aligned with the BRD and include:

- Identifying the most effective platform between Facebook and Instagram
- Analyzing audience engagement and conversion behavior
- Understanding demographic performance patterns
- Tracking budget utilization and measuring ROI
- Detecting trends across months, weeks, and hours

The final dashboard successfully meets these goals and provides several additional insights.

3. In-Scope and Out-of-Scope

In Scope

- Analysis of paid campaigns on Facebook and Instagram
- Event-based performance metrics including impressions, clicks, shares, comments, and purchases
- Demographic and regional performance evaluations

Out of Scope

- Organic interactions
 - Messenger and Audience Network campaigns
 - Non-Meta advertising platforms
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4. KPIs Implemented

KPIs as per BRD

- Impressions
- Clicks
- Shares
- Comments
- Purchases
- Engagements
- CTR
- Engagement Rate
- Conversion Rate
- Purchase Rate
- Total Budget

- Average Budget per Campaign

Additional KPIs Implemented (Beyond BRD)

- ROAS (Return on Ad Spend)
- CPC (Cost Per Click)
- CPM (Cost Per Thousand Impressions)
- Cost per Purchase
- Funnel Drop-off Analysis
- Platform Contribution %
- Age-Group Cost Efficiency
- Ad-Type Effectiveness Score

These additional metrics elevate the analytical depth of the report.

5. Data Model and Transformations

The analytical model was designed using the following tables:

- ads
- ad_events
- users
- Date dimension table

Key Data Preparation Steps

- Cleaning and validating event-level data
- Extracting date-time features such as month, weekday, week number, and hour
- Creating DAX measures for all KPIs and dynamic metrics
- Establishing one-to-many relationships between tables
- Parameter-based measure switching for a more interactive user experience

- Categorization of demographic attributes
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6. Visuals Implemented as per BRD

The following visuals were developed exactly as required in the BRD:

- Donut chart representing gender-wise performance
 - Age group bar chart with dynamic metric selection
 - Map visual displaying country-level performance
 - Calendar heatmap highlighting monthly variations
 - Weekly stacked column chart for ad type comparison
 - Hourly performance area chart
 - Matrix visual summarizing performance by ad type and platform
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7. Additional Visuals Implemented (Beyond BRD)

To enhance the depth of analysis, several additional visuals were designed:

- End-to-end Conversion Funnel (Impression → Click → Engagement → Purchase)
- Platform-level comparison (Facebook vs Instagram)
- KPI summary banner including spend efficiency metrics (CPC, CPM, ROAS)
- Campaign performance table with sortable KPIs
- Dynamic slicers for demographic, regional, and temporal segmentation
- Ad-type ranking visuals based on engagement and purchases

These visuals significantly improve decision-making capability.

8. Analytical Problems Addressed

The report addresses several critical business questions:

Platform Performance Evaluation

Comparison of Facebook and Instagram to determine which delivers better reach, engagement, and conversions.

Audience Segmentation

Identification of high-performing demographics based on gender, age groups, and regions.

Conversion Funnel Assessment

Analysis of user drop-offs at each stage of the funnel to determine where improvements are needed.

Budget Efficiency Measurement

Evaluation of which campaigns justify their spending through ROAS and other cost metrics.

Ad-Type Effectiveness

Comparison of different ad formats to determine which creative style performs best.

Time-Based Optimization

Discovery of peak and low-performance hours, weeks, and months to optimize ad scheduling.

True ROI Calculation

Integration of CPC, CPM, and ROAS to measure the financial effectiveness of campaigns.

9. Key Findings from the Analysis

Platform Insights

- One platform demonstrated stronger reach, while the other achieved higher conversion efficiency.
- ROAS values highlighted clear differences in budget returns across platforms.

Demographic Insights

- Certain gender groups displayed higher engagement and purchase rates.
- Age groups such as 18–24 showed the strongest response to ads.
- Some countries delivered high impressions at lower cost, improving efficiency.

Ad-Type Insights

- Specific ad formats (such as carousel or video) consistently outperformed others.
- Ad-type contributions varied significantly across weekly and hourly windows.

Time-Based Insights

- Peak activity was observed during particular evening hours.
- Weekly trends highlighted inconsistent performance across weeks.
- Monthly trends revealed patterns useful for strategic planning.

Cost and ROI Insights

- CPC and CPM varied widely across demographic segments.
- Certain campaigns showed high spend but low conversions.
- ROAS clearly identified profitable and underperforming campaigns.

10. Business Impact of the Analysis

The dashboard offers several strategic and operational benefits:

Improved Audience Targeting

Clear identification of the most profitable age groups, gender segments, and regions.

Optimized Budget Allocation

Campaigns and demographics with higher ROAS receive prioritized funding.

Enhanced Conversion Rates

Funnel analysis helps reduce drop-offs and improve purchase outcomes.

Stronger Creative Strategy

Insights guide selection of the most effective ad formats for future campaigns.

Efficient Scheduling

Time-based insights assist in publishing ads during high-engagement windows.

Faster Strategic Decisions

The dynamic interactive dashboard reduces analysis time and provides immediate clarity.

Tangible ROI Improvement

By identifying low-performing campaigns and directing budget towards high-performing ones, the overall marketing ROI can significantly increase.

11. Final Outcome

The final Meta Ads Performance Analysis Dashboard:

- Meets all BRD requirements
 - Includes multiple advanced features beyond the documented scope
 - Provides comprehensive insights across platforms, demographics, timelines, and creative formats
 - Supports data-driven marketing decisions
 - Is suitable for presentation, reporting, academic evaluation, and portfolio inclusion
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12. Conclusion

This report provides an end-to-end analytical solution for monitoring and optimizing Meta Ads performance. Through structured data modeling, advanced DAX calculations, clear visualization design, and business-focused insights, the project extends well beyond standard reporting.

The resulting dashboard serves as a complete analytical tool capable of guiding strategic marketing decisions with measurable impact.
